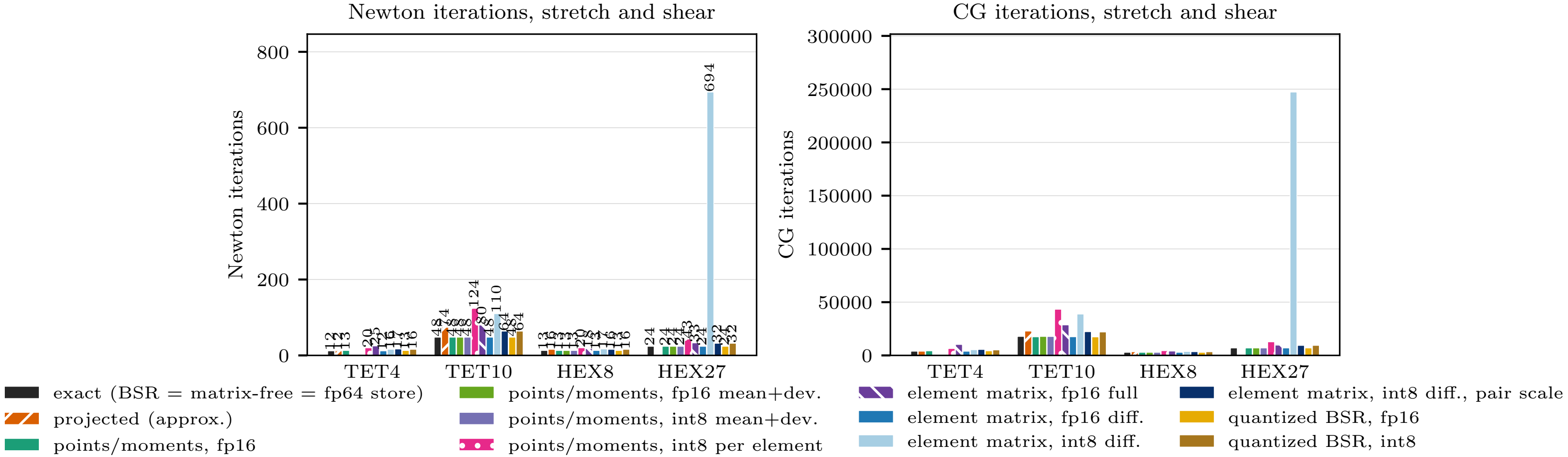
Newton iterations, stretch and shear
Newton iterations
0
200
400
600
800
12
12
13
20
25
12
16
17
13
16
48
74
48
48
48
124
80
48
110
64
48
64
13
16
13
13
13
20
18
13
17
16
13
16
24
24
24
24
43
33
24
694
32
24
32
TET4
TET10
HEX8
HEX27
CG iterations, stretch and shear
CG iterations
0
50000
100000
150000
200000
250000
300000
TET4
TET10
HEX8
HEX27
exact (BSR = matrix-free = fp64 store)
projected (approx.)
points/moments, fp16
points/moments, fp16 mean+dev.
points/moments, int8 mean+dev.
points/moments, int8 per element
element matrix, fp16 full
element matrix, fp16 diff.
element matrix, int8 diff.
element matrix, int8 diff., pair scale
quantized BSR, fp16
quantized BSR, int8